AN IDEAL INSIDE THE JACOBSON RADICAL OF SMALLER COHOMOLOGICAL DIMENSION PULLS THE COHOMOLOGICAL DIMENSION OF A LARGER IDEAL DOWN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The vanishing of local cohomology above the dimension of the support and the bound cd(A, I) \leq dim(A). Author lemma lemma-cd-dimension, whose statement is reproduced verbatim as a context span of this card (source lines 758-764). Gives the a priori bound for the pair of ideals and the vanishing at the non-maximal prime.
- The criterion for the comparison map between two local cohomology modules to be surjective in terms of the vanishing of the local cohomology at the primes of the difference. Author lemma lemma-isomorphism, whose statement is reproduced verbatim as a context span of this card (source lines 485-506). Reduces the vanishing needed for cd(A, I) < d to a surjectivity statement on the top local cohomology of the ring.

Lemma 0.1. *Let A be a Noetherian ring of dimension d. Let $I \subset I' \subset A$ be ideals. If I' is contained in the Jacobson radical of A and $cd(A, I') < d$, then $cd(A, I) < d$.*

Lemma 0.2. *Let $I \subset A$ be a finitely generated ideal of a ring A. If M is a finite A-module, then $H^i_{V(I)}(M) = 0$ for $i > \dim(Supp(M))$. In particular, we have $cd(A, I) \leq \dim(A)$.*

Lemma 0.3. *Let $I \subset I' \subset A$ be finitely generated ideals of a Noetherian ring A. Let M be an A-module. Let $i \geq 0$ be an integer. Consider the map*

$$\Psi : H^i_{V(I')}(M) \to H^i_{V(I)}(M)$$

The following are true:

(1) *if $H^i_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ for all $\mathfrak{p} \in V(I) \setminus V(I')$, then Ψ is surjective,*

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(2) *if* $H^{i-1}_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ *for all* $\mathfrak{p} \in V(I) \setminus V(I')$*, then* Ψ *is injective,*
(3) *if* $H^i_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = H^{i-1}_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ *for all* $\mathfrak{p} \in V(I) \setminus V(I')$*, then* Ψ *is an isomorphism.*

Proof. By Lemma 0.2 we know $\mathrm{cd}(A, I) \leq d$. We will use Lemma 0.3 to show

$$H^d_{V(I')}(A) \to H^d_{V(I)}(A)$$

is surjective which will finish the proof. Pick $\mathfrak{p} \in V(I) \setminus V(I')$. By our assumption on I' we see that $\mathfrak{p}$ is not a maximal ideal of A. Hence $\dim(A_\mathfrak{p}) < d$. Then $H^d_{\mathfrak{p}A_\mathfrak{p}}(A_\mathfrak{p}) = 0$ by Lemma 0.2. □